

What Marvista (by Salesforce Research) is:

A human-AI news reader that guides you through the whole flow — **plan** → **read** → **reflect** — using NLP to (a) fit reading to your **time budget**, (b) surface **question-driven** passages (They use **MixQG** (a T5-based model) to create likely, answerable questions from the article), (c) ask **AI prompts** while you read, and (d) produce an **explainable human-AI summary** at the end → **extractive summarizer: BERT embeddings + K-Means** sentence clustering; select sentences near cluster centroids until the target compression is met..

Marvista: Exploring the Design of a Human-AI Collaborative News Reading Tool

92:3

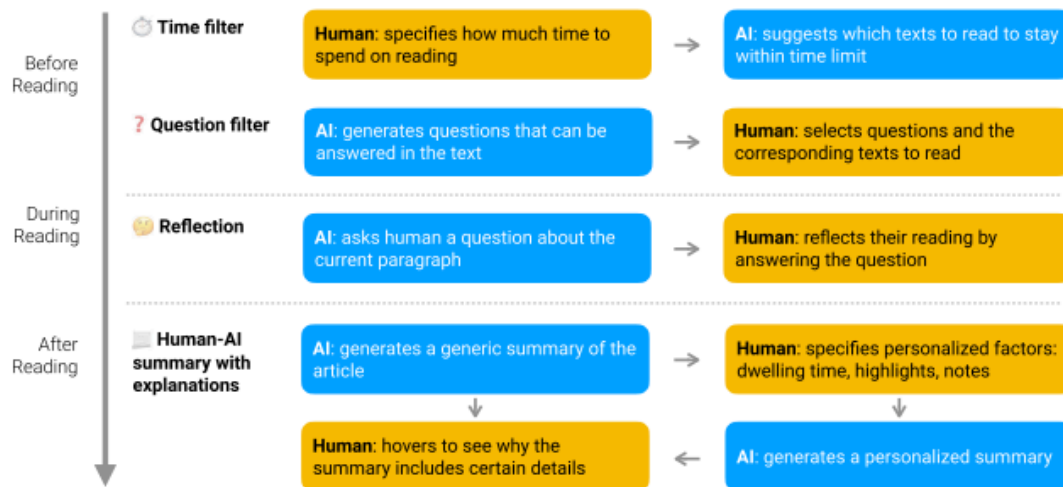


Fig. 1. We explore the design of Marvista—a human-AI collaborative tool that supports the entire process of reading an online article. Before reading: Marvista helps a user plan what to read by filtering text based on how much time one can spend (time filter) and what questions one is interested to find out (question filter); During reading: Marvista helps the user reflect on their understanding of each paragraph with an AI-generated question (reflection); After reading: Marvista generates a human-AI summary that combines AI’s processing of the text, the user’s reading behavior, and user-generated data; further, hovering over a sentence explains which paragraph it is related to and why its information is included in the summary.

CARE (Collaborative AI-Assisted Reading Environment):

it logs reading behavior and plugs into **model-agnostic NLP “assistants”** (QA, summarization, classifiers) via a broker, so you can study/support reading in real time.

Making the logs so that an NLP assistant (QA, summarizer, classifiers) can actually change how people read/comment. Logs enable before/after or A/B comparisons of behavior and outcomes.

Paper Plain CHI 23:

reading UI for medical papers that helps non-experts by:

- 1**

What did the paper want to find out?

This is a review of the evidence and reasons why doctors are using peptides to treat lupus.

see paragraph on page 3

2

What did the paper do?

The next paragraphs tell us about some drugs that are being tested to see if they can help people with lupus.

see paragraph on page 5

3

What were the new treatment(s), if any, this paper looked into?

A peptide is a small molecule made up of amino acids, which are the building blocks of proteins.

see paragraph on page 3

4

What did the paper find?

In the future, scientists will need new drugs that will help people with lupus.

see paragraph on page 18, page 20, page 19

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Conclusion

Better knowledge of the pathogenesis of SLE is expected to enrich the therapeutic armamentarium and facilitate the management of the disease. The use of peptides, specifically designed to target SLE-related epitopes or crucial pathways, may represent a novel fascinating opportunity. Given their good safety profile and immunomodulatory properties, therapeutic peptides could be added to standard of care, and, perhaps, allow the sparing of conventional drugs. In addition, their prescription might be tailored to specific subsets of patients having the highest likelihood of response. Nevertheless, despite the successful results observed in pre-clinical studies, RCTs showed a controversial efficacy profile concerning the use of these compounds in SLE. It is expected that future research, aiming at the amelioration of the physicochemical properties and at the improvement in the management of clinical trials, will bring more encouraging data to innovative therapeutic panorama.

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Expert opinion

The treatment of SLE relies on a conservative approach, combining multiple immunosuppressive agents [8] and, consequently, carrying the risk of unwanted side effects. Unlike other autoimmune diseases, the licensed use of biologics agents which selectively inhibit a specific target has

cytic cells may unleash, through an impaired phagocytosis, the cascade of events leading to type I IFN production, and amplify, through this pathway, inflammation and tissue damage.

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sequence by means of recombinant DNA technology and then modified in order to enhance their pharmacologic properties. Most of therapeutic peptides designed for SLE treatment are synthesized on the basis of immunodominant epitope sequences that are pathogenic in SLE alone. Therefore, they exert an immunomodulatory effect on autoreactive pDC and lymphocytes, without affecting, instead, the immune response against pathogens [26].

To date, no therapeutic peptide has been licensed and marketed for the use in SLE patients. The 21-mer peptide P140 is the only one entering phase III RCTs, and, despite controversial results [36,40,41,43], its development is still ongoing. Edratide, synthesized on the basis of the hCDR1 expressing the major idiotype 16/6 id, showed promising results in preclinical studies [48,50,54,55], but failed to meet the primary endpoint in a phase II RCT [57], with the following interruption of further clinical development. Both the two peptides appeared more effective in anti-dsDNA seropositive patients, and, although data are lacking, a beneficial role may be supposed in those SLE manifestations related to anti-dsDNA antibodies, such as glomerulonephritis.

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Fig. 1. Pictured is the PAPER PLAIN user interface. When a reader opens a paper in PAPER PLAIN, they see a side pane containing a reading guide (1), consisting of key questions the reader might ask of the paper, brief generated plain language answers, and pointers to passages in the paper where the reader can read more. When a reader clicks a question (2), the paper jumps to the passage that provides that answer and shows a paragraph-length plain language answer (3). Plain language summaries can be accessed for any section of the paper by clicking a label next to the section header (4). The reader can view definitions of medical terminology by clicking underlined terms (5).